

DSA0108 - OBJECT ORIENTED PROGRAMMING WITH C++**ASSESSMENT - CO3-AT2-LABORATORY PERFORMANCE****1.STUDENT**

Write a C++ program to create a class for a student with a constructor and a destructor.

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL `simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=constdest`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and a "C++ PROGRAMMING" tag. A user profile dropdown shows "Ragul E" with ID "192571032RD".

On the left sidebar, a dropdown menu is set to "Student", and the section title "STUDENT" is displayed. Below it, the instruction reads: "Write a C++ program to create a class for a student with a constructor and a destructor."

The main editor area contains the following C++ code:

```
1#include <iostream>
2#include <string>
3using namespace std;
4
5class Student {
6public:
7    string* name;
8    int rollNumber;
9
10    Student(string n, int r) {
11        name = new string(n);
12        rollNumber = r;
13        cout << "Constructor: Memory allocated for " << *name << endl;
14    }
15
16    void display() {
17        cout << "Name: " << *name << ", Roll Number: " << rollNumber << endl;
18    }
19};
20
21int main() {
```

On the right side, the input field contains the text "ragul" and "101".

Below the code editor, the "OUTPUT" section shows the execution results:

```
Constructor: Memory allocated for KGF
Name: KGF, Roll Number: 101
```

2.BOOK

Write a C++ program to create a class for a book with a constructor and a destructor.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar indicates the URL: `simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=constdest`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and a "C++ PROGRAMMING" tag. A user profile for "Ragul E" is visible in the top right corner.

On the left sidebar, under "QUESTIONS 5/5 completed", the "Book" question is selected. The question text reads: "Write a C++ program to create a class for a book with a constructor and a destructor."

The main area is the "PROGRAM EDITOR" for C++. It contains the following code:

```
1 #include <iostream>
2 using namespace std;
3 class Book
4 {
5 private:
6     string title;
7     string author;
8 public:
9     Book(string t, string a)
10    {
11        title = t;
12        author = a;
13        cout << "\nConstructor called" << endl;
14    }
15    void display()
16    {
17        cout << "\nBook Title : " << title << endl;
18        cout << "Book Author : " << author << endl;
19    }
20    ~Book()
21    {
```

On the right, the "INPUT" section shows the user input: "OOPS" and "RAGUL".

OUTPUT

```
Enter Book Title: Enter Book Author:
Constructor called

Book Title   : OOPS
Book Author  : RAGUL

Destructor called
```

3.RECTANGLE

Write a C++ program to create a class for a rectangle with a constructor and a destructor.

The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and a user profile for "Ragul E" with ID "192571032RD". The main content area is divided into three sections: "QUESTIONS" (5/5 completed), "PROGRAM EDITOR", and "INPUT". The "QUESTIONS" section shows a dropdown menu with "Rectangle" selected and a description: "Write a C++ program to create a class for a rectangle with a constructor and a destructor." The "PROGRAM EDITOR" section shows a C++ code editor with the following code:

```
1 #include <iostream>
2 using namespace std;
3
4 class Rectangle
5 {
6 private:
7     float length;
8     float width;
9
10 public:
11     Rectangle(float l, float w)
12     {
13         length = l;
14         width = w;
15         cout << "\nConstructor called!" << endl;
16     }
17     void display()
18     {
19         cout << "\nLength : " << length << endl;
20         cout << "Width : " << width << endl;
21         cout << "Area : " << length * width << endl;
22     }
23 }
```

The "INPUT" section shows the input values: 4 and 5.

OUTPUT

```
Enter Length: Enter Width:
Constructor called!

Length : 4
Width  : 5
Area   : 20

Destructor called! Object destroyed.
```

4.CAR

Write a C++ program to create a class for a car with a constructor and a destructor.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar indicates the URL: `simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=constest`. The page header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and a "C++ PROGRAMMING" tag. A user profile for "Ragul E" is visible in the top right corner.

On the left side, under the "QUESTIONS" tab, the question "Car" is selected. The question text reads: "Write a C++ program to create a class for a car with a constructor and a destructor."

The "PROGRAM EDITOR" tab is active, showing a C++ program. The code is as follows:

```
1 #include <iostream>
2 using namespace std;
3
4 class Car
5 {
6 private:
7     string brand;
8     string model;
9
10 public:
11     Car(string b, string m)
12     {
13         brand = b;
14         model = m;
15         cout << "\nConstructor called!" << endl;
16     }
17     void display()
18     {
19         cout << "\nCar Brand : " << brand << endl;
20         cout << "Car Model : " << model << endl;
21     }
22 }
```

Buttons for "Run" and "Save" are located above the code editor. The "INPUT" tab on the right shows the input values: "bmw" and "X100".

OUTPUT

```
Enter Car Brand: Enter Car Model:
Constructor called!

Car Brand : bmw
Car Model : X100
```

5.BANK

Write a C++ program to create a class for a bank account with a constructor and a destructor.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar indicates the URL: `simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=constdest`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "Ragul E 192571032RD".

On the left, the "QUESTIONS" panel shows "5/5 completed" and a dropdown menu with "Bank" selected. Below it, the "BANK" section contains the instruction: "Write a C++ program to create a class for a bank account with a constructor and a destructor."

The "PROGRAM EDITOR" panel shows the following C++ code:

```
1 #include <iostream>
2 using namespace std;
3
4 class Bank
5 {
6 private:
7     string name;
8     int accountNo;
9     float balance;
10
11 public:
12     Bank(string n, int a, float b)
13     {
14         name = n;
15         accountNo = a;
16         balance = b;
17         cout << "\nConstructor called!" << endl;
18     }
19     void display()
20     {
21         cout << "\nAccount Holder : " << name << endl;
```

The "INPUT" panel on the right shows the input values: "Ragul", "278456976320", and "200".

OUTPUT

```
Enter Account Holder Name: Enter Account Number: Enter Balance:
Constructor called!
```

```
Account Holder : Ragul
Account Number : 2147483647
Balance        : 4.59037e-41
```

```
Destructor called! Object destroyed.
```